

HEALTH DATA ANALYTICS

Hospital Performance, Patient Demographics & Treatment Cost Report

A Business Intelligence Review of the Power BI "Health Data" Dashboard

By - Ayansh Singh

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1. Executive Summary

This report summarizes the findings of the “Health Data” Power BI dashboard, a three-page analytical model covering Patient Demographics, Key Trends, and Treatment & Cost across a network of ten partner hospitals. The dashboard tracks admissions, average length of stay (LOS), billing, diagnostic outcomes, and medical condition distribution, with interactive filters for month, year, hospital, medical condition, and insurance provider.

Across the full 2022 reporting year, the network admitted 11,017 patients, supported by 9,339 attending doctors and 400 rooms/bedspaces, at an average billing amount of \$25,514 per patient and an average length of stay of 15.36 days. The average patient age across the dataset is approximately 52 years.

Three findings stand out for leadership attention:

- Extended-stay admissions (15+ days) generate the largest share of total billing across every admission type (Elective, Emergency, Urgent), signaling that length-of-stay management is the single largest lever for cost control.
- Diagnostic testing shows a high abnormal-result rate (54% abnormal vs. 11% normal in 2022), which should be validated against clinical protocols to rule out over-testing or referral bias.
- Admission volume is heavily weekday-weighted (approximately 71% weekday vs. 29% weekend in 2022), which has direct implications for weekday staffing and bed-turnover planning.

2. Dashboard Overview & Scope

The Power BI file (Health_Data_Project.pbix) is organized into three navigable report pages, each sharing a common set of six headline KPI cards — Admitted Patients, Rooms/Bedspace, Avg. Billing Amount, Doctors, Avg. LOS (Days), and Avg. Age — alongside a Select Month and Select Year slicer that drives every visual on the page.

2.1 Report Pages

- Patients Demographics — admissions by blood group, age group, and medical condition; diagnostic test result outcomes (Normal / Inconclusive / Abnormal); and hospital-level condition breakdown.
- Key Trends — admissions by day type (weekday vs. weekend) and by LOS bucket; a monthly trend chart switchable between Admitted Patients, Avg. LOS, and Avg. Billing Amount; and a day-of-week by month calendar matrix.
- Treatment & Cost — total billing by LOS bucket and admission type; total billing by hospital or insurance provider (toggle); and a medical condition by medication billing/volume matrix.

2.2 Interactivity

All three pages support drill-down via Select Hospital Name, Select Medical Condition, and a Hospital / Insurance Provider toggle, allowing the same KPI framework to be interrogated at the network level, the single-hospital level, or the single-condition level.

2.3 Reference Views

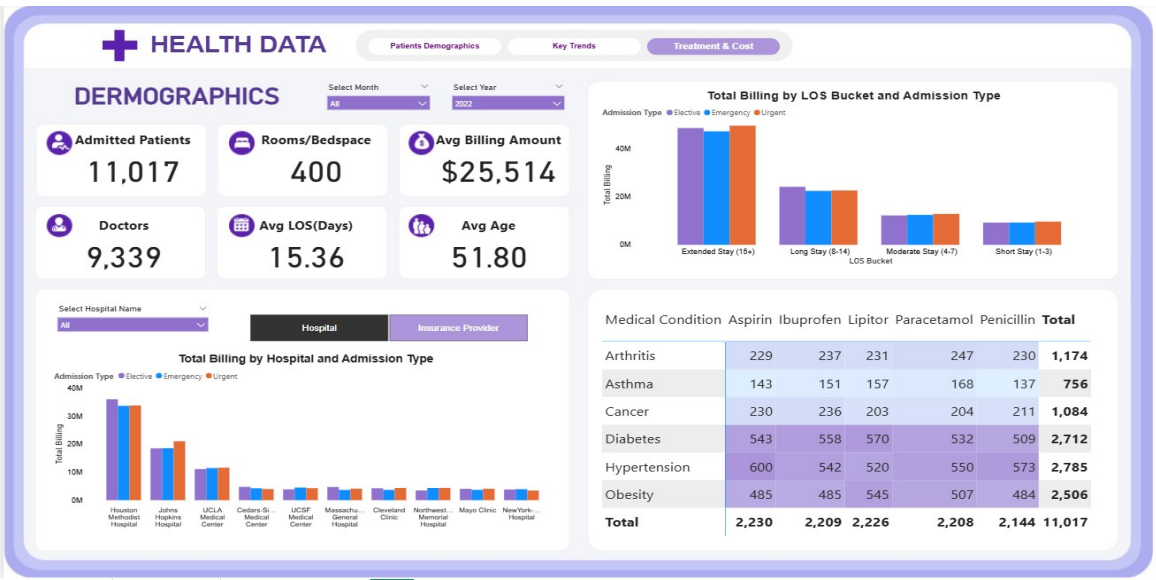


Figure 1. Treatment & Cost page — full network, all months, 2022.

3. Patient Demographics

The Demographics page provides the clinical and population profile of admitted patients. At the full-network, all-year 2022 view, the following headline metrics apply:

Metric	Value
Admitted Patients	11,017
Rooms / Bedspace	400
Avg. Billing Amount	\$25,514
Doctors	9,339
Avg. LOS (Days)	15.36
Avg. Age	51.80

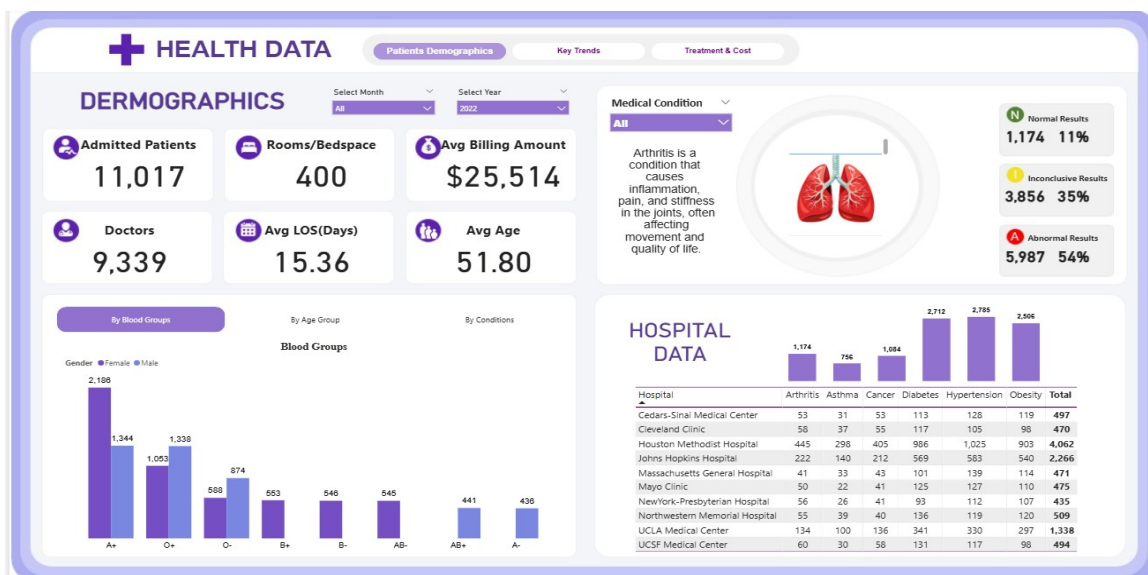


Figure 2. Patients Demographics page — full network, all months, 2022.

3.1 Diagnostic Test Outcomes (2022, All Conditions)

Result Category	Count	Share of Tests
Normal Results	1,174	11%
Inconclusive Results	3,856	35%
Abnormal Results	5,987	54%

More than half of recorded diagnostic tests return an abnormal result, and only about one in nine returns a clean normal result. This pattern warrants review — either the patient population is heavily skewed toward confirmed illness (consistent with an inpatient, already-admitted cohort), or testing/coding protocols merit a closer audit.

3.2 Blood Group Distribution (by Gender)

Blood Group	Female	Male
A+	2,186	1,344
O+	1,053	1,338
O-	588	874
B+	553	—
B-	546	—
AB-	545	—
AB+	—	441
A-	—	436

A+ is the most common blood type in the admitted population, followed by O+ and O-. This distribution is useful input for blood bank provisioning and transfusion-readiness planning at the network level.

3.3 Admissions & Conditions by Hospital (2022)

Hospital	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity	Total
Houston Methodist Hospital	445	298	405	986	1,025	903	4,062
Johns Hopkins Hospital	222	140	212	569	583	540	2,266
UCLA Medical Center	134	100	136	341	330	297	1,338
Cedars-Sinai Medical Center	53	31	53	113	128	119	497
Northwestern Memorial Hospital	55	39	40	136	119	120	509
NewYork-Presbyterian Hospital	56	26	41	93	112	107	435
Mayo Clinic	50	22	41	125	127	110	475
UCSF Medical Center	60	30	58	131	117	98	494
Massachusetts General Hospital	41	33	43	101	139	114	471
Cleveland Clinic	58	37	55	117	105	98	470

Houston Methodist Hospital and Johns Hopkins Hospital together account for roughly 57% of total admissions across all six tracked conditions, making them the two largest contributors to network patient volume. Diabetes and hypertension are the two most prevalent conditions network-wide (2,712 and 2,785 recorded cases respectively in 2022), followed closely by obesity (2,506 cases).

3.4 Single-Condition Drill-Down Example: Cancer, April 2023

Filtering to April 2023 and the Cancer condition isolates a cohort of 110 admitted patients, 96 rooms/bedspace utilized, an average billing amount of \$23,812, and an average age of 52.33 — with 100% of diagnostic results in this filtered view classified as Abnormal, as expected for a confirmed-cancer cohort.

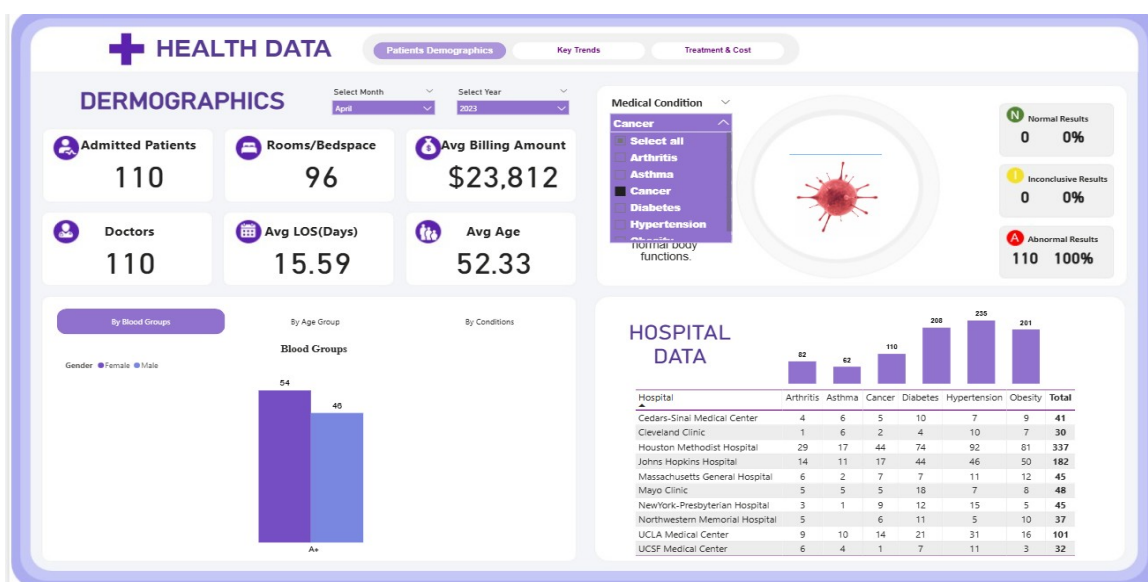


Figure 3. Patients Demographics page — Cancer condition, April 2023.

4. Key Trends

The Key Trends page focuses on the temporal and operational shape of admissions: when patients are admitted, how long they stay, and how those patterns move month to month.

4.1 Admissions by Day Type (2022, Full Network)

Day Type	Admitted Patients	Share
Weekday	~8,000	71.29%
Weekend	~3,000	28.71%

4.2 Admissions by Length-of-Stay Bucket (2022, Full Network)

LOS Bucket	Admitted Patients	Share
Extended Stay (15+ days)	~6,000	52.51%
Long Stay (8–14 days)	~3,000	24.45%
Moderate Stay (4–7 days)	~1,000	13.06%
Short Stay (1–3 days)	~1,000	9.98%

More than half of all admitted patients fall into the Extended Stay (15+ days) bucket. Combined with Long Stay, over three-quarters of the admitted population (76.96%) stays 8 or more days — a distribution more consistent with a high-acuity, tertiary-care network than a general short-stay hospital population.

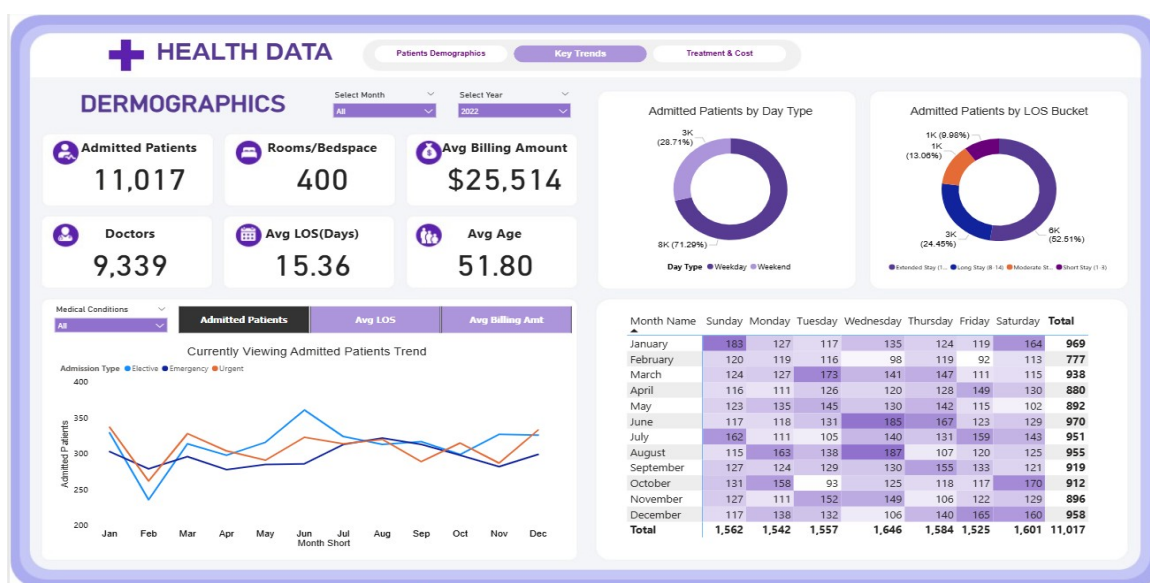


Figure 4. Key Trends page — full network, all months, 2022.

4.3 Monthly Admission Trend (2022, Full Network)

Month	Total Admitted Patients
January	969

Month	Total Admitted Patients
February	777
March	938
April	880
May	892
June	970
July	951
August	955
September	919
October	912
November	896
December	958
Total (2022)	11,017

Monthly admissions are relatively stable across 2022, ranging from a low of 777 (February) to a high of 970 (June), with no strong seasonal spike — suggesting admission volume is driven more by referral and bed-capacity patterns than by seasonal illness cycles at the aggregate level.

4.4 Single-Condition Drill-Down Example: Hypertension, 2023

Filtering to the Hypertension condition for 2023 shows 2,793 total admissions across the year, with April 2023 alone contributing 235 admissions, an average LOS of 15.64 days, and an average billing amount of \$25,416 — both figures tracking close to the network-wide averages, indicating hypertension patients do not disproportionately drive cost or length of stay relative to the general admitted population.

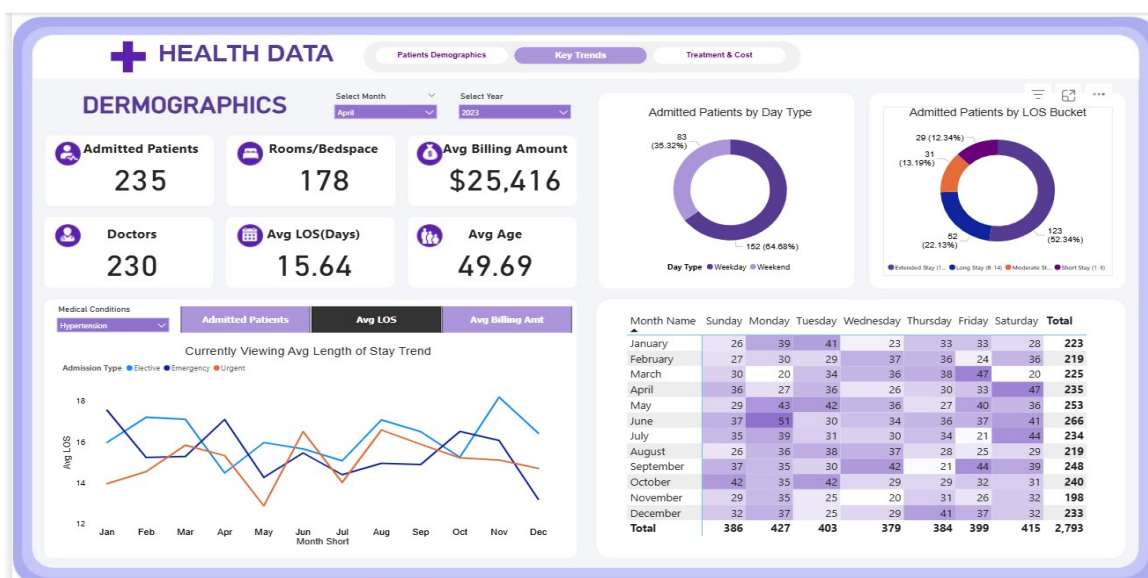


Figure 5. Key Trends page — Hypertension condition, April 2023.

5. Treatment & Cost

The Treatment & Cost page translates the clinical and operational picture into financial terms, breaking down total billing by length-of-stay bucket, admission type, hospital, insurance provider, and medical condition/medication combination.

5.1 Billing by LOS Bucket and Admission Type (2022, Full Network)

Total billing rises sharply with length of stay. The Extended Stay (15+ days) bucket generates roughly four times the billing of the Short Stay (1–3) bucket across all three admission types (Elective, Emergency, and Urgent), which are broadly balanced with one another within each bucket. This confirms that length-of-stay reduction — not admission-type mix — is the primary cost lever available to hospital operations.

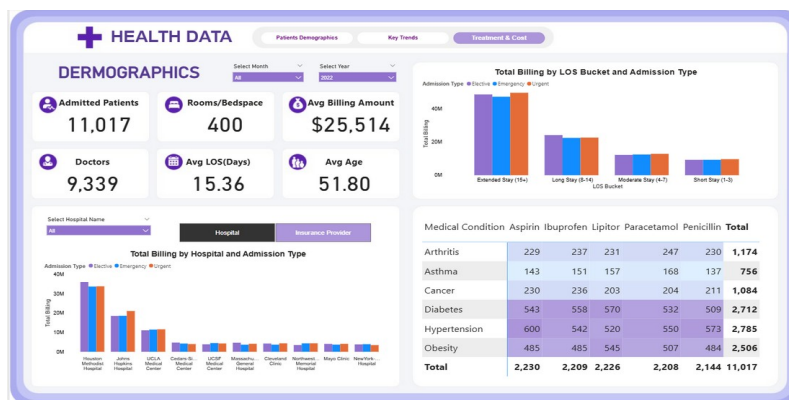


Figure 6. Total Billing by LOS Bucket and Admission Type, 2022.

5.2 Billing by Hospital and Insurance Provider

The dashboard allows toggling the billing chart between a Hospital view and an Insurance Provider view. In the single-hospital example below (Johns Hopkins Hospital, February 2021), Medicare is the leading payer by total billing, followed by UnitedHealthCare, with Cigna and Aetna contributing smaller shares — a payer mix that is broadly consistent with an older admitted-patient population (average age ~52).

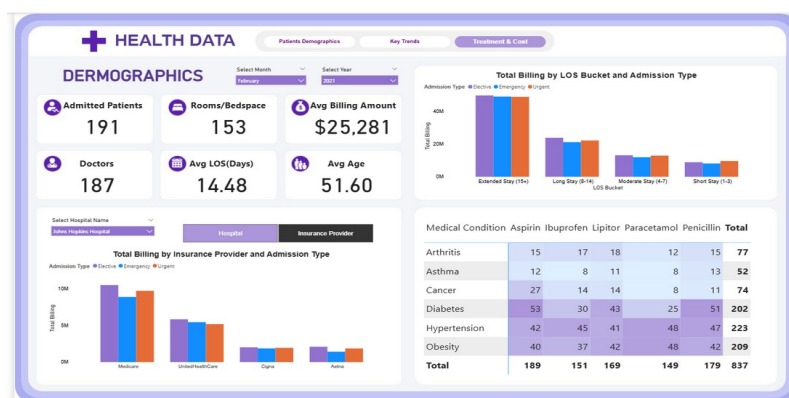


Figure 7. Total Billing by Insurance Provider — Johns Hopkins Hospital, February 2021.

5.3 Medical Condition × Medication Matrix (2022, Full Network)

Condition	Aspirin	Ibuprofen	Lipitor	Paracetamol	Penicillin	Total
Arthritis	229	237	231	247	230	1,174
Asthma	143	151	157	168	137	756
Cancer	230	236	203	204	211	1,084
Diabetes	543	558	570	532	509	2,712
Hypertension	600	542	520	550	573	2,785
Obesity	485	485	545	507	484	2,506
Total	2,230	2,209	2,226	2,208	2,144	11,017

Medication volume is spread almost evenly across the five tracked drugs (each accounting for 19–20% of total recorded doses), indicating no single medication dominates the treatment protocol mix. Hypertension and Diabetes are the two largest condition-medication combinations by volume, together representing roughly half of all recorded condition-medication records.

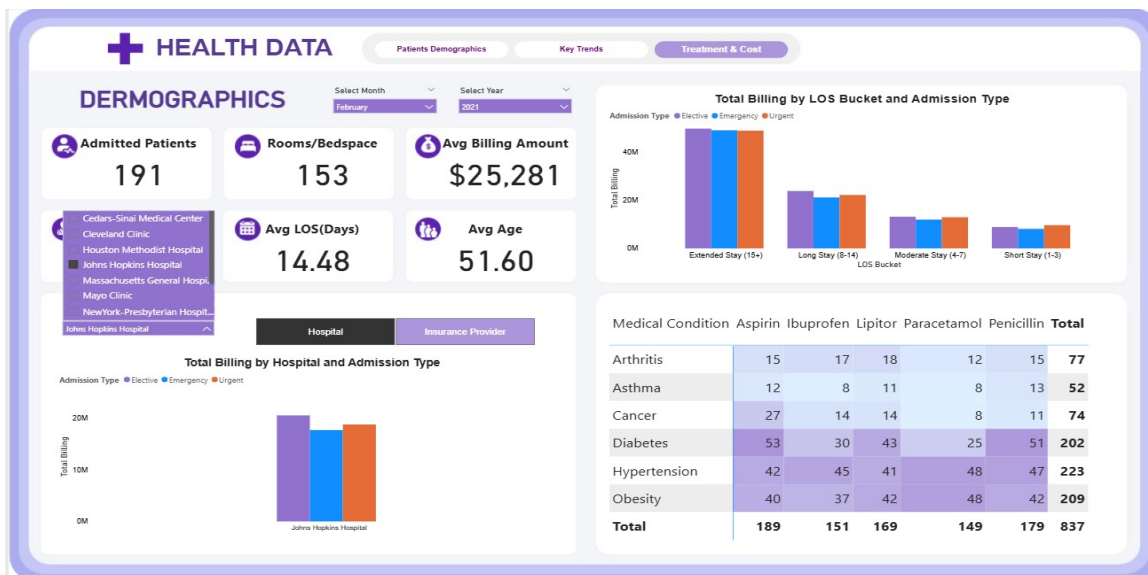


Figure 8. Treatment & Cost page — hospital selector detail, Johns Hopkins Hospital, February 2021.

6. Key Insights

- Length of stay is the dominant cost driver. Extended-stay patients (15+ days) make up over half of admissions and generate proportionally the highest billing across every admission type and hospital. LOS reduction initiatives (discharge planning, care coordination) offer the clearest path to cost containment.
- Admission volume is weekday-concentrated (~71%), which has direct staffing implications — weekday bed turnover and discharge capacity should be prioritized over weekend capacity expansion.
- Two hospitals (Houston Methodist and Johns Hopkins) account for the majority of network-wide condition volume, suggesting these sites may be operating at or near capacity relative to peers and could be candidates for load-balancing or capacity investment.

- Diabetes and Hypertension are the leading chronic conditions network-wide by both admission volume and medication volume, reinforcing the case for dedicated chronic-disease management programs.
- The high abnormal-result rate (54%) merits clinical review to confirm it reflects appropriate case-mix (an already-admitted, higher-acuity population) rather than a testing or documentation issue.
- Monthly admission volume is stable year-round with no pronounced seasonality, supporting steady-state (rather than seasonally flexed) staffing and bed-capacity planning.

7. Recommendations

- Launch a targeted length-of-stay reduction program for the Extended Stay (15+ day) cohort, starting with the two highest-volume hospitals, given its outsized share of both patient days and total billing.
- Review weekday staffing models and discharge workflows to relieve the weekday admission concentration and improve bed-turnover velocity.
- Expand chronic-disease management (diabetes and hypertension) programs, which together represent roughly half of all condition-medication activity network-wide.
- Commission a clinical audit of diagnostic testing protocols to validate the 54% abnormal-result rate against expected case-mix for an inpatient population.
- Use the blood-group distribution data to inform blood bank provisioning, prioritizing A+ and O-type inventory given their outsized share of the admitted population.
- Extend the dashboard with a Length-of-Stay vs. Cost-per-Day efficiency metric to separate hospitals that are simply higher-volume from those that are genuinely less cost-efficient per patient-day.